

CSCI 5244: Homework set 1

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- Given a state of a qubit $|\psi\rangle = a|0\rangle + b|1\rangle$, where $a, b \in \mathbb{C}$ and $|a|^2 + |b|^2 = 1$, and recall that states $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$

- In 2-dimensional complex space with basis $|0\rangle, |1\rangle$, write down the matrix form of operator $A = |+\rangle\langle+| + \frac{1}{2}|-\rangle\langle-|$ and verify A is an observable (i.e., self-adjoint).

First, the column vector of $|0\rangle$ can be written as $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|1\rangle$ as $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

Solving for $|+\rangle$:

$$\frac{1}{\sqrt{2}} \left(\begin{pmatrix} 0 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Solving for $|-\rangle$:

$$\frac{1}{\sqrt{2}} \left(\begin{pmatrix} 0 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right) = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

Corresponding $\langle+|$ and $\langle-|$ are just the complex conjugate of their respective kets in row vector form.

The complex conjugate to both of these are represented as follows:

$$\begin{aligned} \langle 0| &= \frac{1}{\sqrt{2}} (1 \quad 1) \\ \langle 1| &= \frac{1}{\sqrt{2}} (1 \quad -1) \end{aligned}$$

Solving first operation $|+\rangle\langle+|$:

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \cdot \frac{1}{\sqrt{2}} (1 \quad 1) = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

Solving operation $|-\rangle\langle-|$:

$$\frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix} \cdot \frac{1}{\sqrt{2}} (1 \quad -1) = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

Matrix form of operator A:

$$\begin{aligned} A &= \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \frac{1}{2} \left(\frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \right) \\ &= \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \frac{1}{4} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \\ &= \frac{1}{4} \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix} \end{aligned}$$

Matrix A is self-adjoint iff $A^\dagger = A$ where $A^\dagger$ is the conjugate transpose:

$$A = \frac{1}{4} \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$$

Switch off-diagonal entries:

$$\frac{1}{4} \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$$

Since we are not dealing with any imaginary numbers, no conjugation is needed. Thus proves that $A^\dagger = A$ making it self-adjoint, and making it a valid observable.

- (b) If you measure the observable A on the state $|\psi\rangle$, what are the possible measurement outcomes and associated probabilities?

The measurement of observable A "snaps" onto one of A 's eigenvalues $|n\rangle$ such that:

$$Prob(a_n) = ||E_n|\psi\rangle||^2 = \langle\psi|E_n|\psi\rangle \text{ (from the slides)}$$

The observable A is already in the form $A = |+\rangle\langle+| + \frac{1}{2}|-\rangle\langle-|$, so the expected value of measurement can be written as:

$$A = \sum_n \lambda_n |v_n\rangle\langle v_n|$$

With possible outcomes being λ_n with $Prob(\lambda_n) = |\langle v_n|\psi\rangle|^2$.

Already we can see that the two potential eigenvalues are 1 in front of $|+\rangle\langle+|$ and $\frac{1}{2}$ in front of $|-\rangle\langle-|$

The first probability $Prob(1)$ can be calculated as such:

$$\begin{aligned} \langle+| &= \frac{1}{\sqrt{2}}(\langle 0| + \langle 1|) = \frac{1}{\sqrt{2}}((1 \ 0) + (0 \ 1)) = \frac{1}{\sqrt{2}}(1 \ 1) \\ \langle+|\psi\rangle &= \frac{1}{\sqrt{2}}(1 \ 1) \begin{pmatrix} a \\ b \end{pmatrix} = \frac{1}{\sqrt{2}}(a + b) \end{aligned}$$

$Prob(1) = |\langle+|\psi\rangle|^2$ which is the squared magnitude of $\frac{1}{\sqrt{2}}(a + b)$. And the absolute square of a complex number z is defined as $|z| = zz^*$ where z^* is the complex conjugate.

$$\begin{aligned} Prob(1) &= \frac{1}{2}(|a|^2 + |b|^2 + ab^* + a^*b) \\ &= \frac{1}{2}(1 + ab^* + a^*b) \\ &= \frac{1}{2}(1 + 2Re(ab^*)) \quad a^*b \text{ is the complex conjugate of } ab^* \\ &= \frac{1}{2} + Re(ab^*) \end{aligned}$$

The second probability $Prob(\frac{1}{2})$ can be calculated like so:

$$\begin{aligned} \langle-| &= \frac{1}{\sqrt{2}}(\langle 0| - \langle 1|) = \frac{1}{\sqrt{2}}(1 \ -1) \\ \langle-|\psi\rangle &= \frac{1}{\sqrt{2}}(1 \ -1) \begin{pmatrix} a \\ b \end{pmatrix} = \frac{1}{\sqrt{2}}(a - b) \end{aligned}$$

Using the same logic from Prob(1) where $\langle -|\psi\rangle$ is just the squared magnitude of $\frac{1}{\sqrt{2}}(a-b)$, and the absolute square of a complex number z is defined as $|z| = zz^*$ where z^* is the complex conjugate, we can write out the following:

$$\begin{aligned} \text{Prob}(\frac{1}{2}) &= \frac{1}{2}(|a|^2 + |b|^2 - ab^* - a^*b) \\ &= \frac{1}{2}(1 - (ab^* + a^*b)) \\ &= \frac{1}{2}(1 - 2\text{Re}(ab^*)) \quad a^*b \text{ is the complex conjugate of } ab^* \\ &= \frac{1}{2} - \text{Re}(ab^*) \end{aligned}$$

As a sanity check, if we add these two probabilities together, we should get 1:

$$\frac{1}{2} + \text{Re}(ab^*) + \frac{1}{2} - \text{Re}(ab^*) = 1$$

2. Consider a state of two qubits $|\psi\rangle = \frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle - \frac{i}{\sqrt{2}}|11\rangle$.

- (a) If you were to label the basis of 4-dimensional complex space in the following order: $|00\rangle|01\rangle|10\rangle|11\rangle$ what are the vector forms of $|\psi\rangle$ and $\langle\psi|$?

For $|\psi\rangle$ the basis is defined by the coefficients which can be written in column vector form as such:

$$|\psi\rangle = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ 0 \\ -\frac{i}{\sqrt{2}} \end{pmatrix}$$

For $\langle\psi|$, this can be represented as the complex conjugate of the column vector rewritten as a row vector:

$$\langle\psi| = (\frac{1}{2} \quad \frac{1}{2} \quad 0 \quad \frac{i}{\sqrt{2}})$$

- (b) If you were to label the basis of 4-dimensional complex space in the following order: $|00\rangle|10\rangle|01\rangle|11\rangle$ what are the vector forms of $|\psi\rangle$ and $\langle\psi|$?

$$\begin{aligned} |\psi\rangle &= \begin{pmatrix} \frac{1}{2} \\ 0 \\ \frac{1}{2} \\ -\frac{i}{\sqrt{2}} \end{pmatrix} \\ \langle\psi| &= (\frac{1}{2} \quad 0 \quad \frac{1}{2} \quad \frac{i}{\sqrt{2}}) \end{aligned}$$

- (c) If you were to label the basis of 4-dimensional complex space in the following order: $|00\rangle|01\rangle|10\rangle|11\rangle$ what is the matrix form of density operator $\rho = |\psi\rangle\langle\psi|$?
 ρ is a 4×4 matrix formed by the outer product of $|\psi\rangle\langle\psi|$.

$$\begin{aligned}
|\psi\rangle\langle\psi| &= \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ 0 \\ -\frac{i}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 & -\frac{i}{\sqrt{2}} \end{pmatrix} \\
&= \begin{pmatrix} \frac{1}{4} & \frac{1}{4} & 0 & \frac{i}{2\sqrt{2}} \\ \frac{1}{4} & \frac{1}{4} & 0 & \frac{i}{2\sqrt{2}} \\ 0 & 0 & 0 & 0 \\ -\frac{i}{2\sqrt{2}} & -\frac{i}{2\sqrt{2}} & 0 & \frac{1}{2} \end{pmatrix}
\end{aligned}$$

(d) Is $|\psi\rangle$ a separable state or entangled state?

$|\psi\rangle$ is separable if it can be written as $(a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle)$ for some a, b, c, d .

For $|\psi\rangle = \frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle - \frac{i}{\sqrt{2}}|11\rangle$, the coefficients are as follows:

$$ac = \frac{1}{2}, \quad ad = \frac{1}{2}, \quad bc = 0, \quad bd = -\frac{i}{\sqrt{2}}$$

With case $b = 0$, the equation $bd = -\frac{i}{\sqrt{2}} \neq 0$.

With case $c = 0$, $ac = \frac{1}{2} \neq 0$.

This state does not follow $(a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle)$, and therefore this state is entangled and cannot be separated.

3. Recall the quantum gate notation and diagrams, and consider the following circuit (not shown in my homework, but assume that it is there).

(a) What is the output of the above circuit?

Applying Hadamard gate to the first qubit:

$$\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

Apply CNOT gate:

$$\begin{aligned}
&= \frac{1}{2}[(|0\rangle + |1\rangle) \otimes (|0\rangle + |1\rangle)] \\
&= \frac{1}{2}(|00\rangle + |01\rangle + |01\rangle + |11\rangle) \\
&= \frac{1}{2}(|00\rangle + |01\rangle + |11\rangle + |10\rangle) \text{ CNOT applied}
\end{aligned}$$

Apply phase change $e^{i\pi/4}$:

$$\frac{1}{2}(|00\rangle + |10\rangle + |01\rangle + e^{i\pi/4}|11\rangle)$$

(b) We measure the second qubit on the computational basis $\{|0\rangle, |1\rangle\}$ at the end of computation. After the measurement, what are the possible states of the first qubit, and the associated probabilities?

The equation is:

$$\frac{1}{\sqrt{2}}(|0\rangle|+\rangle + |1\rangle \frac{1}{\sqrt{2}}(|0\rangle + e^{i\pi/4}|1\rangle))$$

For qubit 2 = 0:

$$\frac{1}{2}|00\rangle + \frac{1}{2}|10\rangle = \frac{1}{2}(|0\rangle + |1\rangle) \otimes |0\rangle$$

For qubit 1 = 0:

$$\frac{1}{2}|01\rangle + \frac{1}{2}e^{i\pi/4}|11\rangle = \frac{1}{2}(|0\rangle + e^{i\pi/4}|1\rangle) \otimes |1\rangle$$

The probability of an outcome = squared length ($\langle\psi| = \sum |c_i|^2$)

For qubit 2 = 0:

$$|\frac{1}{2}|^2 + |\frac{1}{2}|^2 = \frac{1}{2}$$

For qubit 2 = 1:

$$\begin{aligned} & |\frac{1}{2}|^2 + |e^{i\pi/4}\frac{1}{2}|^2 \\ &= \frac{1}{4} + \sqrt{\cos(\pi/4)^2 + \sin(\pi/4)^2} \\ &= \frac{1}{4} + \frac{1}{4} = \frac{1}{2} \end{aligned}$$

Normalizing the branches:

$$\begin{aligned} k^2 \cdot \frac{1}{2} &= 1 \\ k^2 &= 2 \\ k &= \sqrt{2} \\ \frac{1}{\sqrt{2}}(|0\rangle + e^{i\pi/4}|1\rangle) \end{aligned}$$

Apply to qubit 2 = 0:

$$\sqrt{2}(\frac{1}{2}|0\rangle + \frac{1}{2}|1\rangle) = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

Apply to qubit 2 = 1:

$$\begin{aligned} & \sqrt{2}(\frac{1}{2}|0\rangle + \frac{1}{2}e^{i\pi/4}|1\rangle) \\ &= \frac{1}{\sqrt{2}}(|0\rangle + e^{i\pi/4}|1\rangle) \end{aligned}$$

Measuring qubit 2 to get the outcome of 0: probability $\frac{1}{2}$, and qubit 1 collapses to $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$

Measuring qubit 2 to get the outcome of 1: probability $\frac{1}{2}$, and qubit 1 collapses to $\frac{1}{\sqrt{2}}(|0\rangle + e^{i\pi/4}|1\rangle)$